

Cloud Computing Lab Manual Submission

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Courses : MCA SEM 2

Subject : Cloud Computing

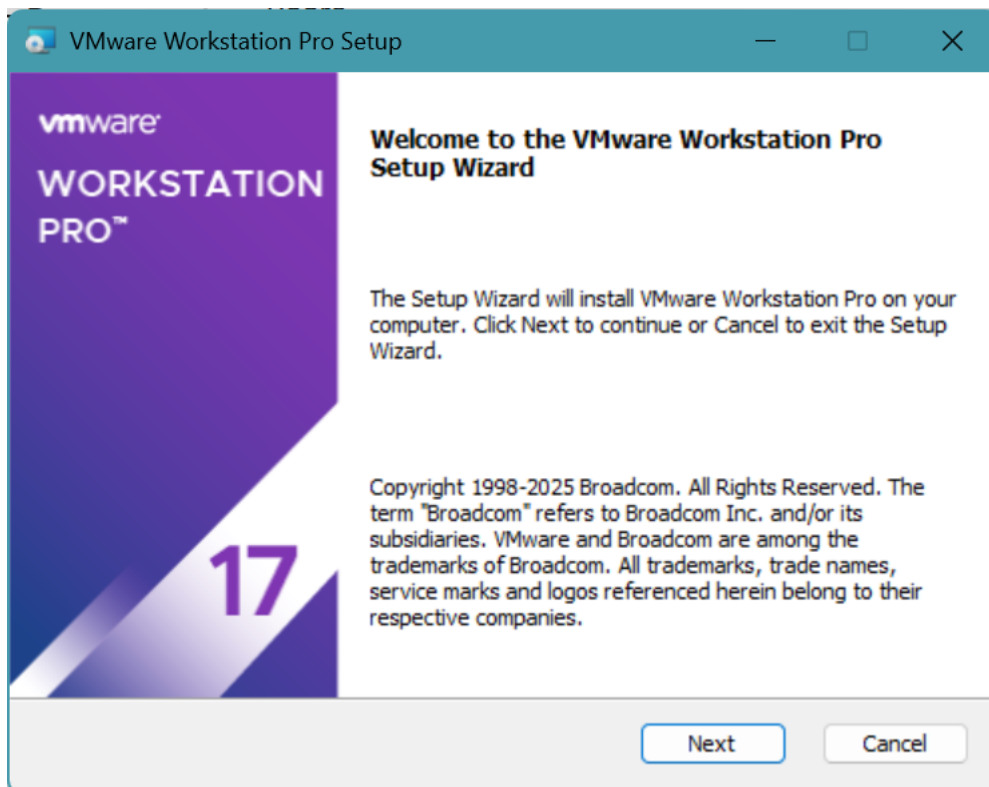
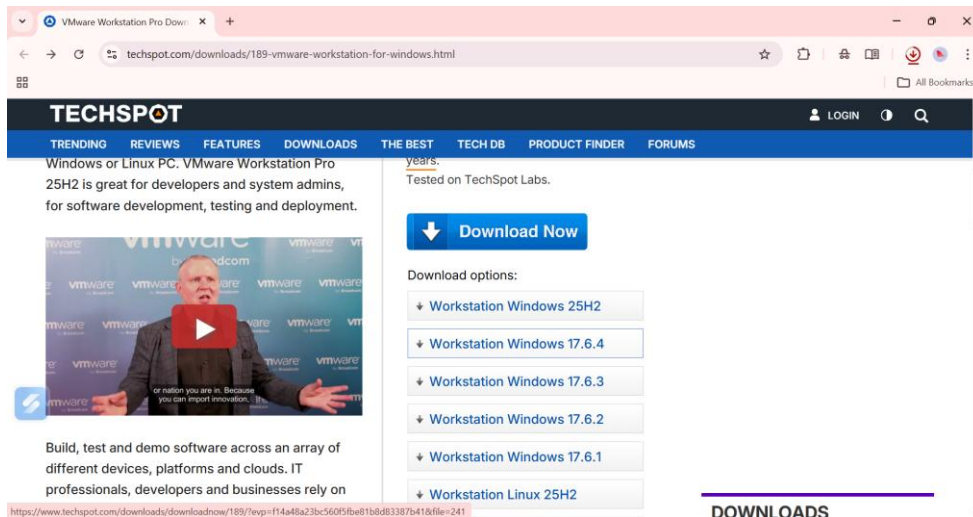
Practical 1 : Introduction to VMware workstation and it's installation Process

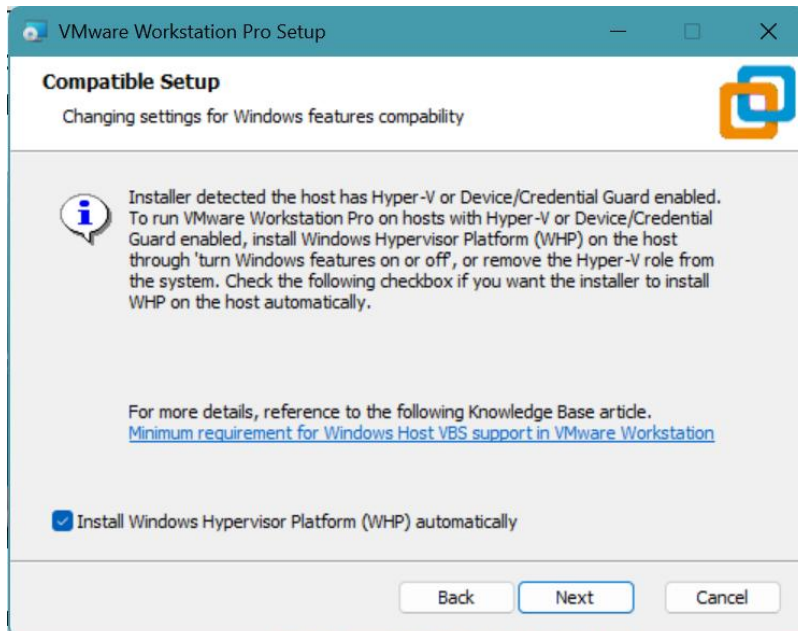
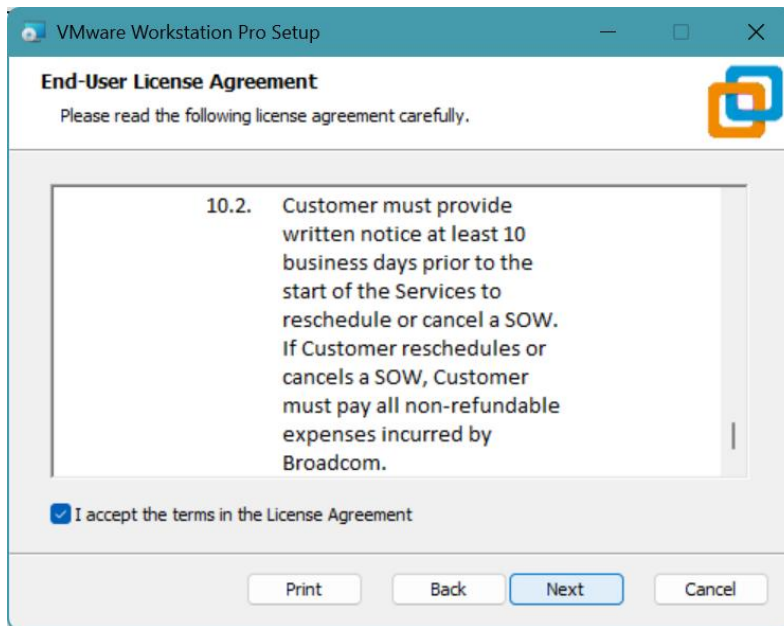
Step 1: Open a web browser and navigate to:

<https://www.techspot.com/downloads/189-vmware-workstation-for-windows.html>

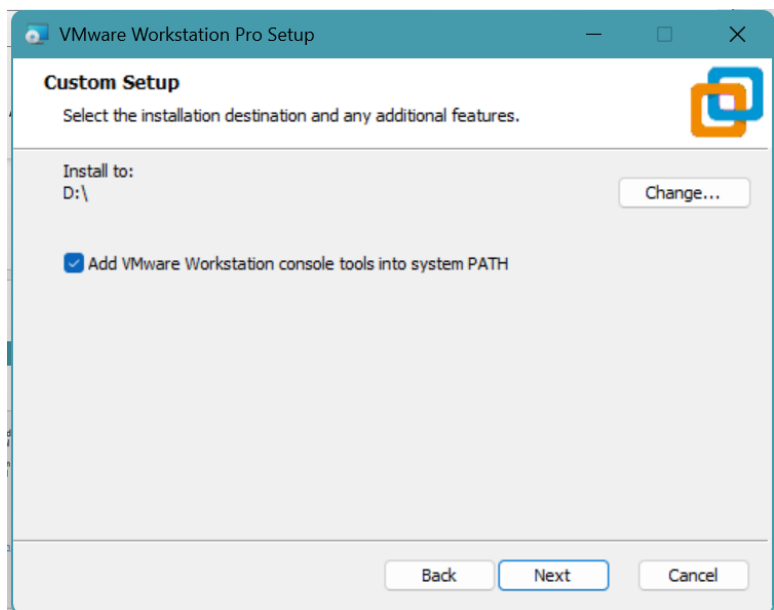
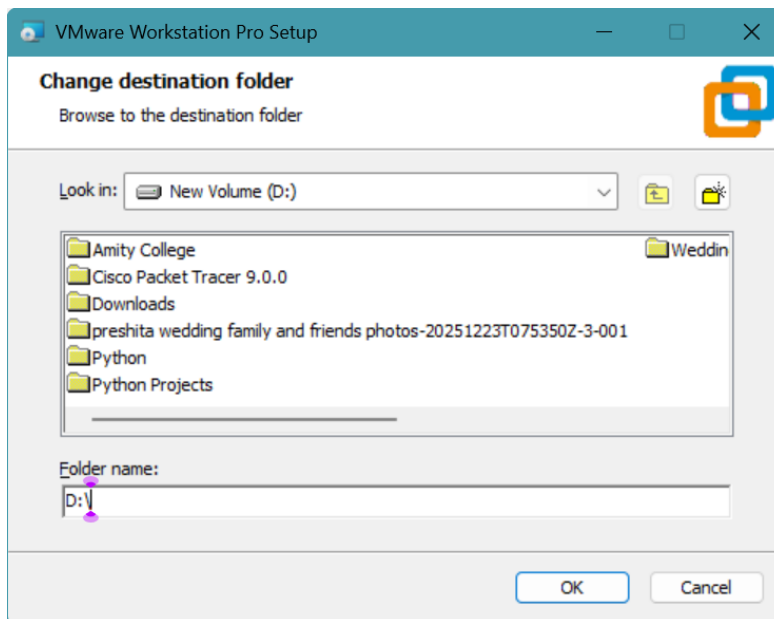
Step 2 : On the TechSpot page, locate the Download Now button and click it.

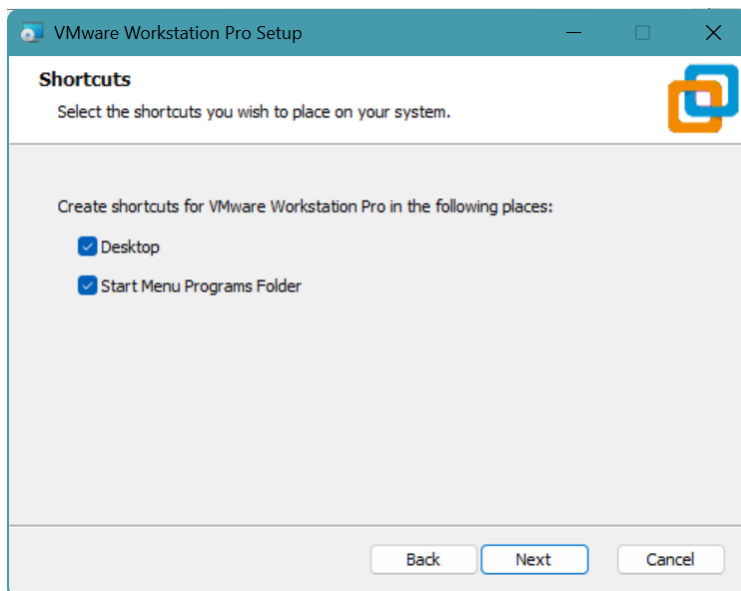
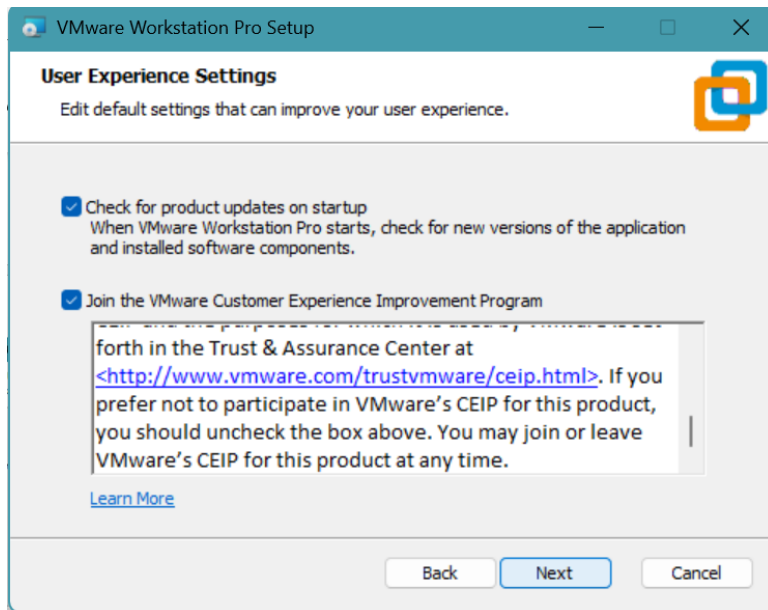
Step 3: From the list of download options, select Workstation Windows 25H2 (or the latest available version for Windows).

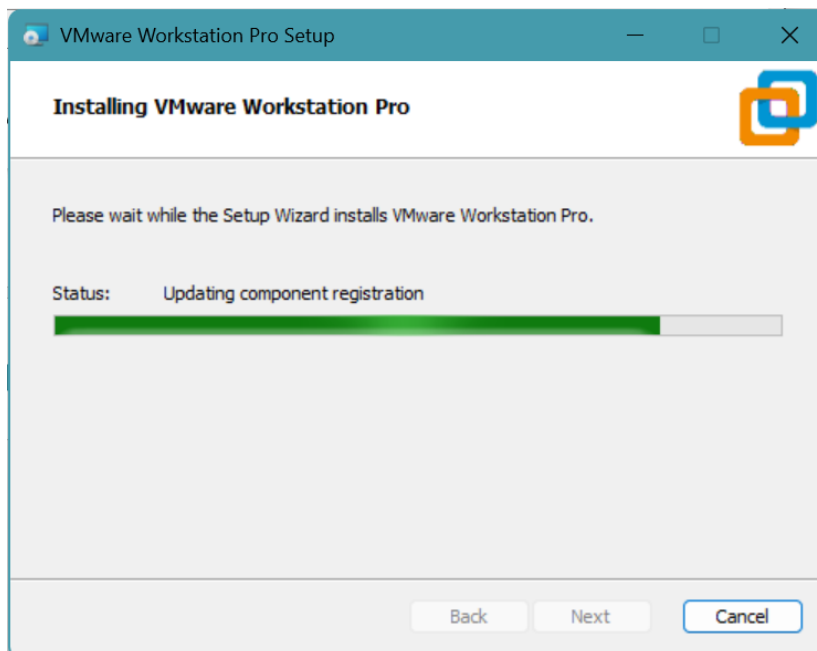
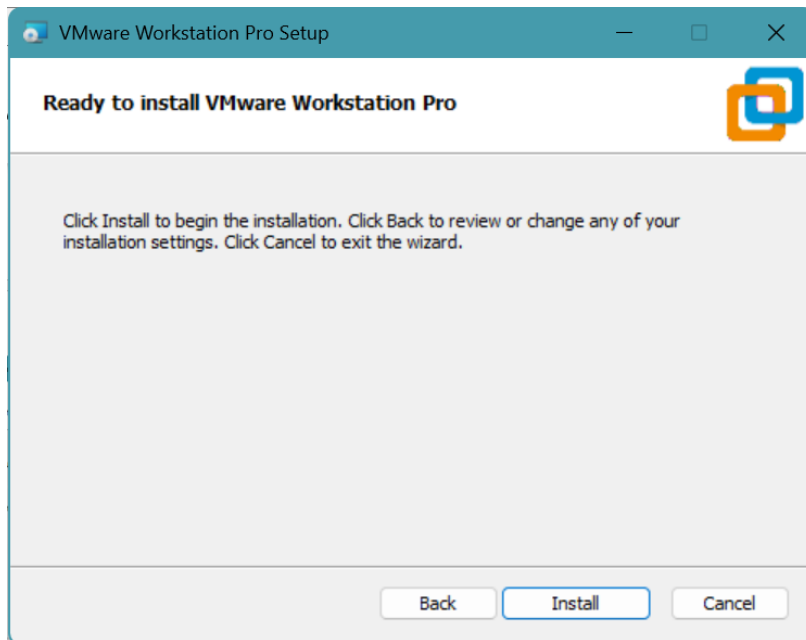




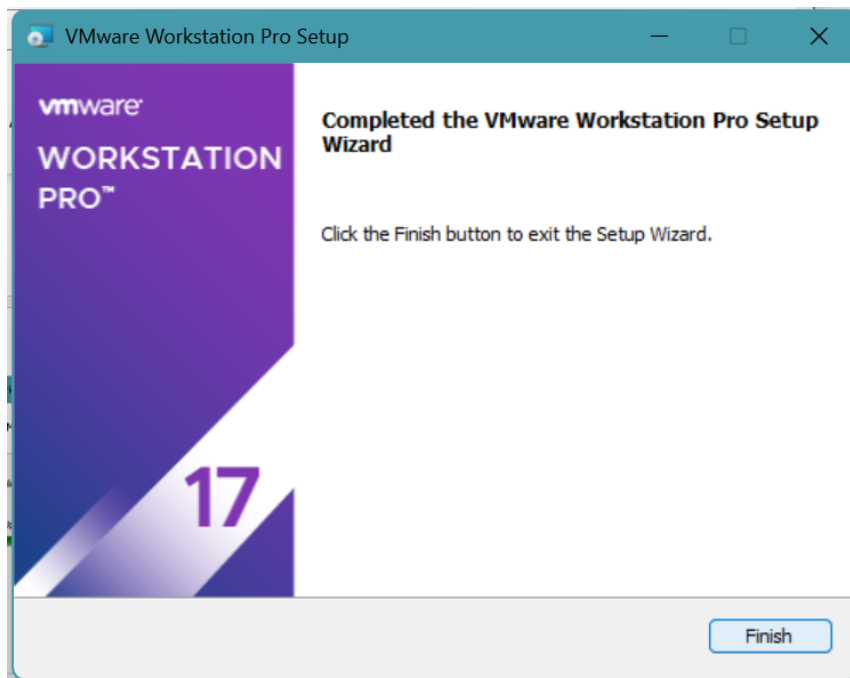
Step 4 : Save the installer file (e.g., VMware-workstation-full-17.x.x.exe) to a known location on your computer.







Step 5 : Once installation finishes, the Completed the VMware Workstation Pro Setup Wizard screen appears. Click Finish to exit the wizard.

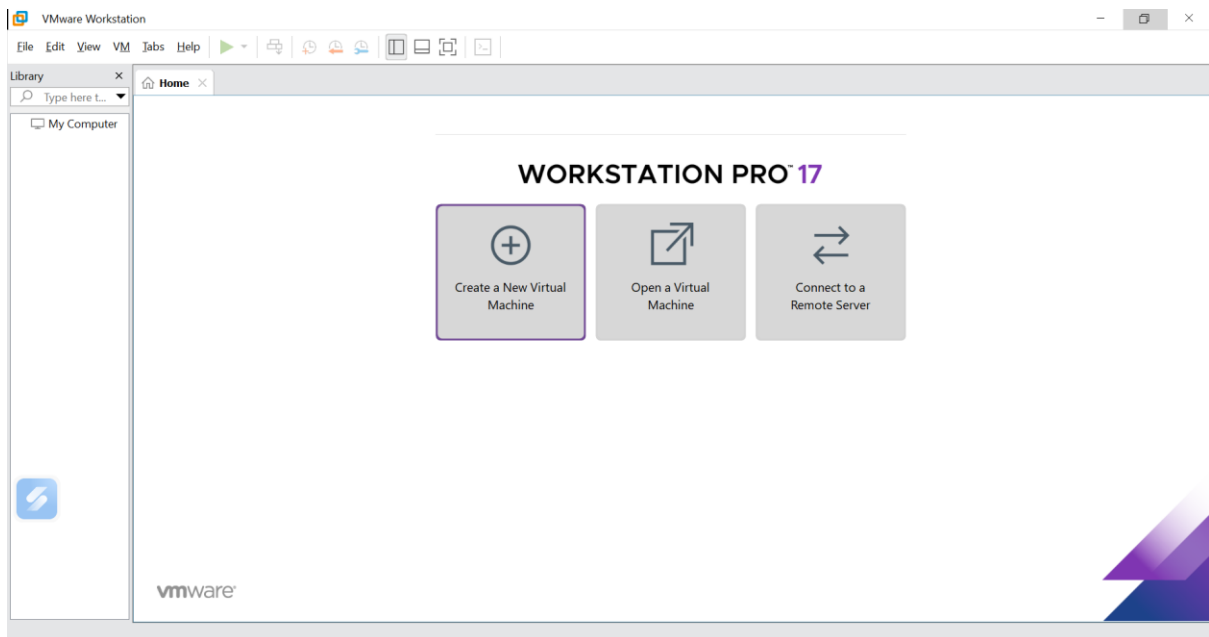


Practical 2 : Basic setting of VMware to install Virtual Machine in VMware Workstation

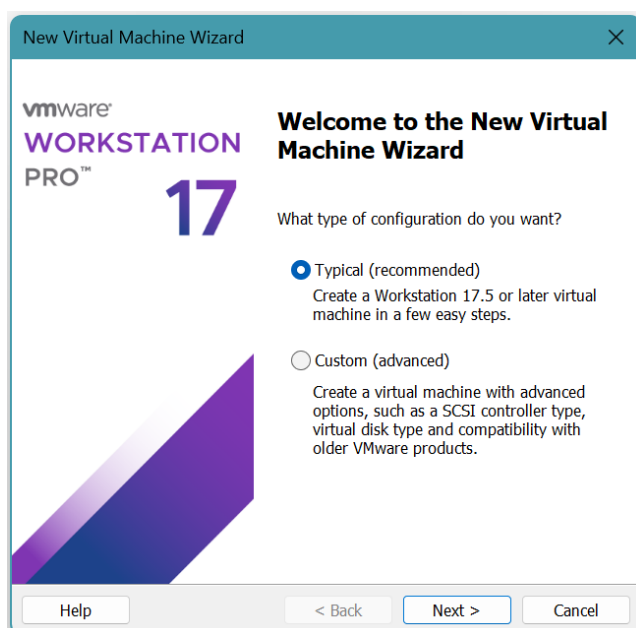
Step 1: Launch VMware Workstation Pro 17 from the Desktop or Start Menu.

Step 2: The VMware Workstation home screen will appear showing three options: Create a New Virtual Machine, Open a Virtual Machine, and Connect to a Remote Server.

Step 3: Click Create a New Virtual Machine. The New Virtual Machine Wizard will open.



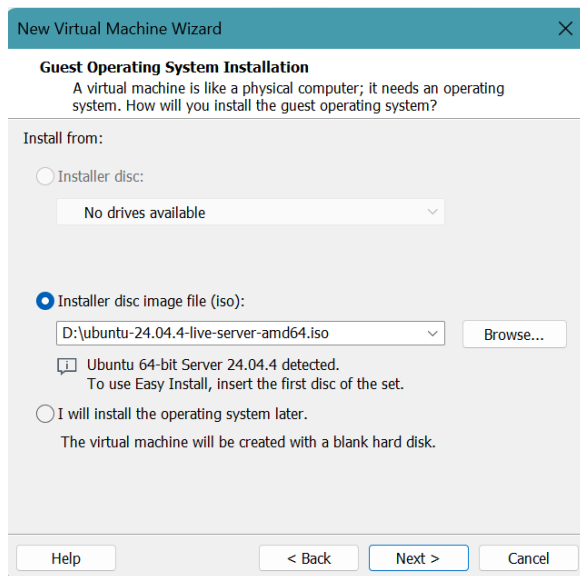
Step 4: Select Typical (recommended) and click Next.



Step 5 : On the Guest Operating System Installation screen, select Installer disc image file (iso)

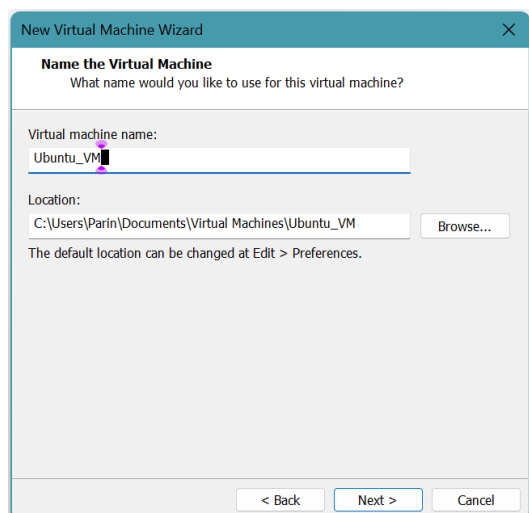
Step 6: Click Browse and navigate to the location of your Ubuntu ISO file (e.g., D:\ubuntu-24.04.4-live-server-amd64.iso).

Step 7 : VMware will automatically detect the OS: Ubuntu 64-bit Server 24.04.4 detected. Click Next.



Step 8: In the Virtual machine name field, enter a name for your VM (e.g., Ubuntu_VM).

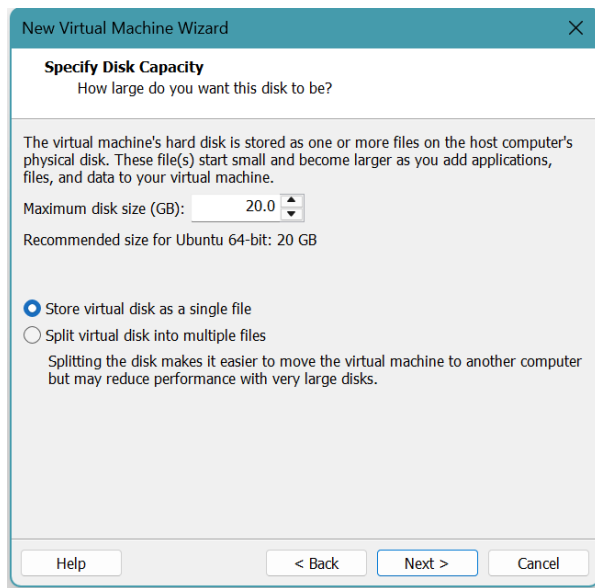
Step 9 : The Location field shows where the VM files will be stored (e.g., C:\Users\YourName\Documents\Virtual Machines\Ubuntu_VM). You may change this by clicking Browse. Click Next.



Step 10: On the Specify Disk Capacity screen, set the Maximum disk size (GB) to 20.0 GB (recommended for Ubuntu 64-bit).

Step 11: Select Store virtual disk as a single file for better performance. Click Next.

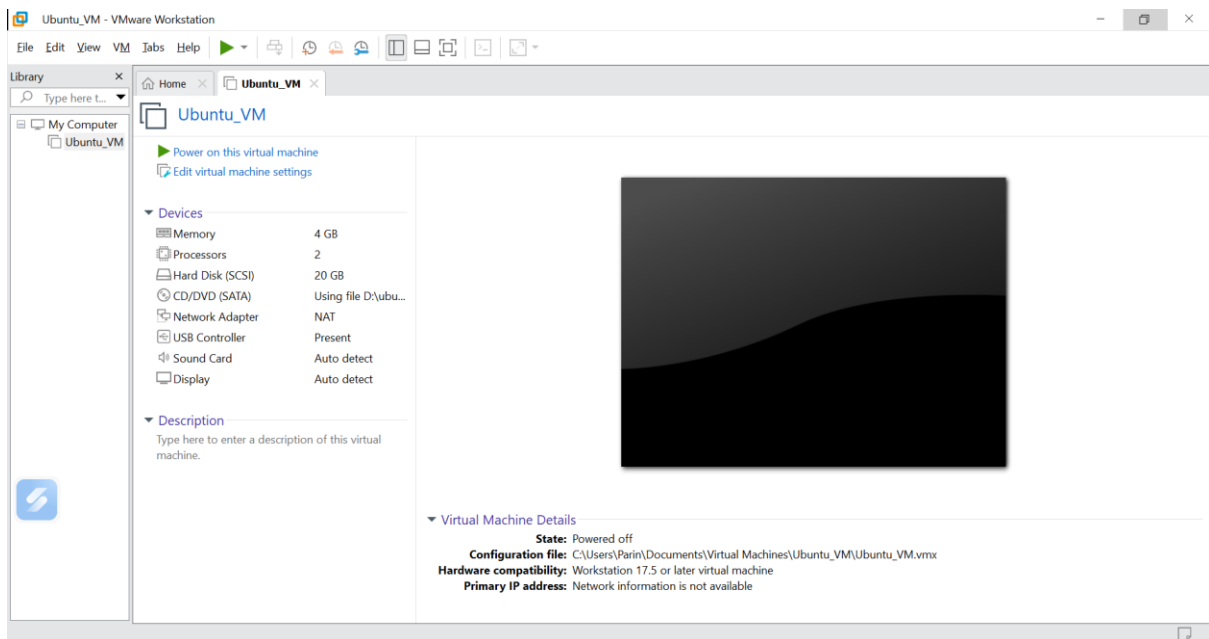
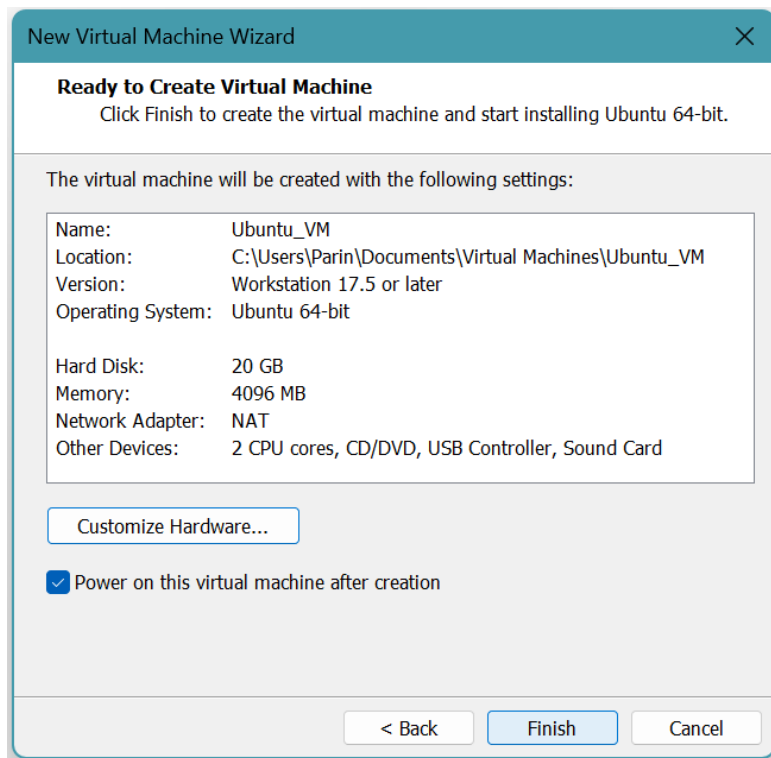
Narayan 32



Step 12: The Ready to Create Virtual Machine screen displays a summary of your VM configuration: Name: Ubuntu_VM, Version: Workstation 17.5 or later, Operating System: Ubuntu 64-bit, Hard Disk: 20 GB, Memory: 4096 MB, Network Adapter: NAT, Other Devices: 2 CPU cores, CD/DVD, USB Controller, Sound Card.

Step 13: Ensure the checkbox Power on this virtual machine after creation is checked if you want to start the VM immediately.

Step 14: Click Finish. The virtual machine Ubuntu_VM will now appear in the VMware library panel.



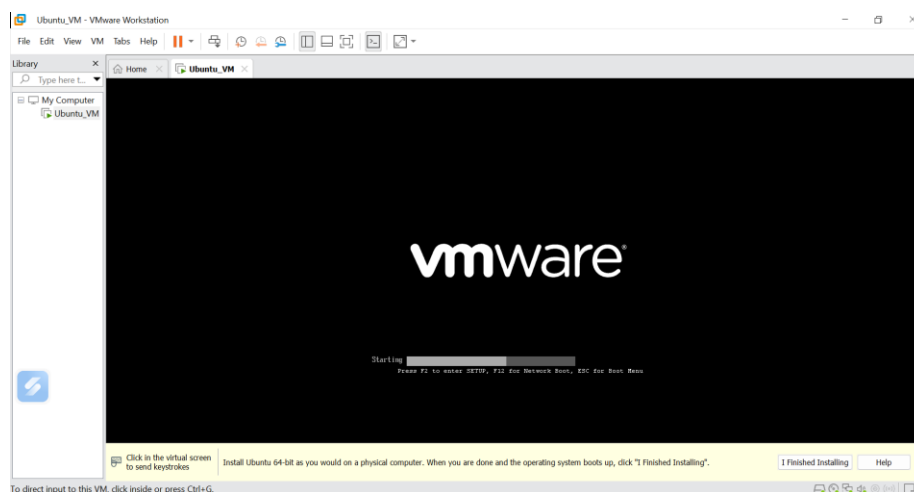
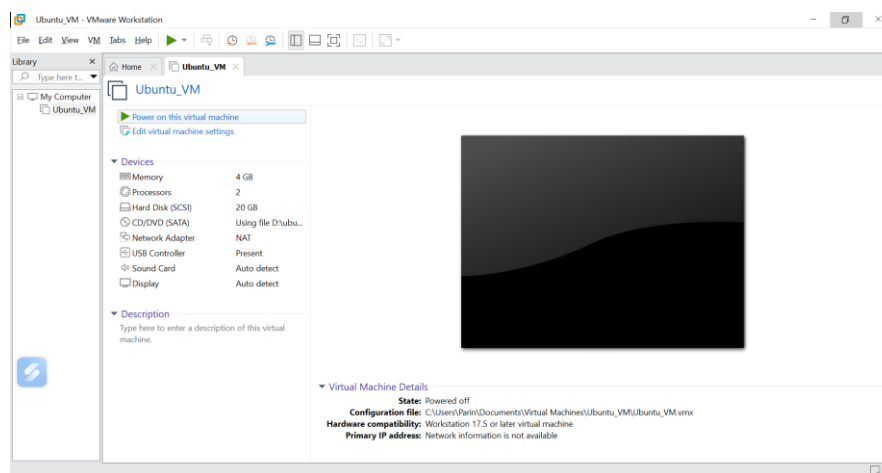
Practical 3 : Installation of Hypervisor and VM in VMware Workstation

Step 1: In the VMware library panel on the left, select Ubuntu_VM.

Step 2: In the main panel, click Power on this virtual machine. Alternatively, you can click the green Play button in the toolbar.

Step 3: The VM screen will appear and show the VMware logo along with the boot message: Starting... Press F2 to enter SETUP, F12 for Network Boot, ESC for Boot Menu.

Step 4: The system will begin booting and loading Ubuntu from the ISO. You will see a stream of system messages in the console as services initialize

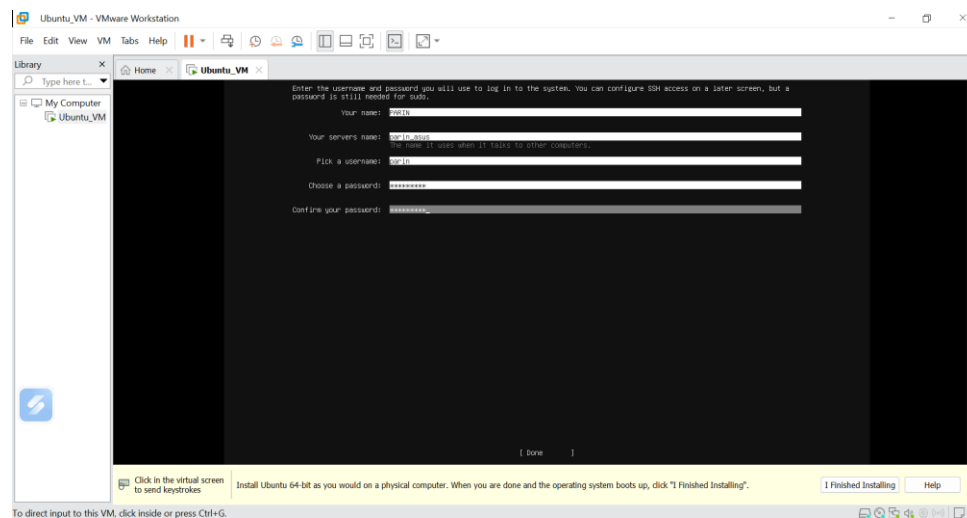
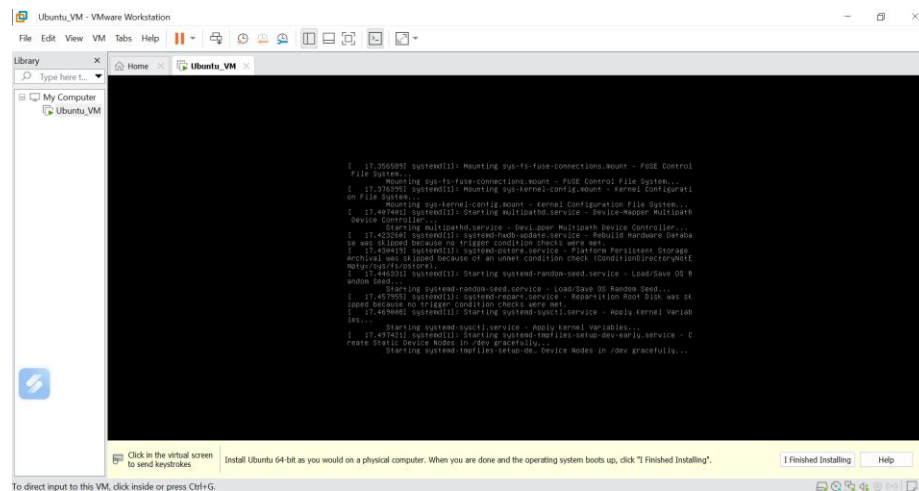


Step 5 : Click inside the VM window to direct keyboard and mouse input to the virtual machine.

Step 6: Follow the on-screen Ubuntu Server installation prompts. When asked to configure the system profile, enter the following: Your name (e.g., Admin), Your server's name (hostname), Your username (e.g., parin), A password, and Confirm the password.

Step 7: Press Done to continue through each section of the installer.

Step 8: Continue through the remaining Ubuntu installation steps (storage configuration, network setup, SSH options, etc.) accepting the defaults unless specific changes are required.

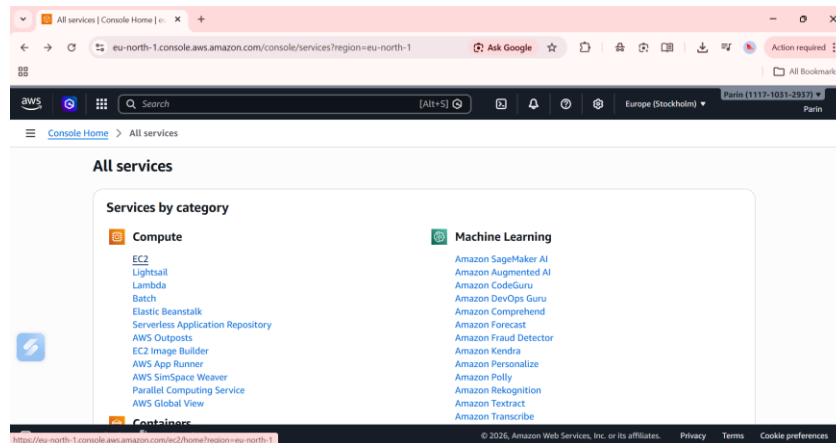


Practical : 5 : Concept and Lab of elastic Compute cloud in AWS

Step 1: Open a web browser and go to: <https://aws.amazon.com> and sign in to your AWS account. Ensure you are signed in under the correct account

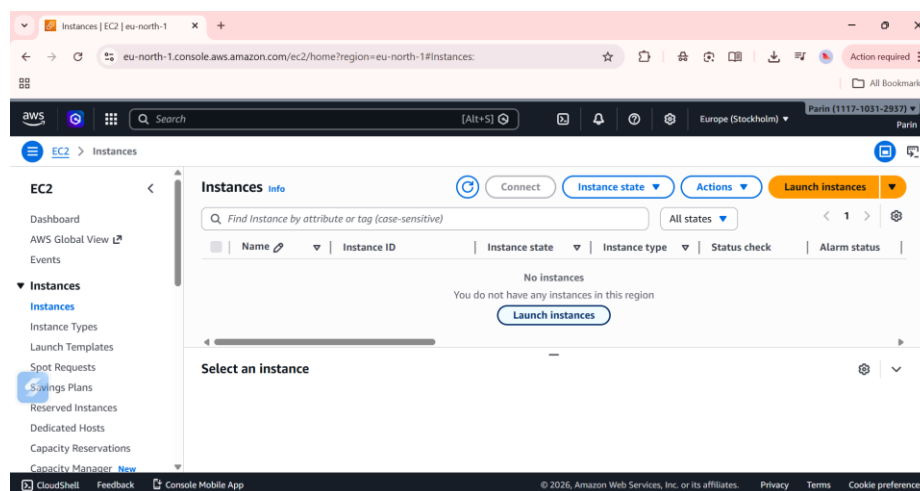
Step 2: Once logged in, click the grid icon (Services) in the top navigation bar, or use the Search bar and type EC2.

Step 3: On the All Services page, locate the Compute section and click EC2

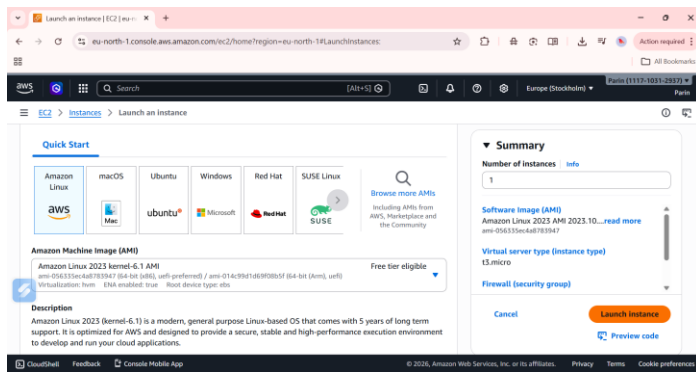


Step 4: You will be directed to the EC2 Dashboard. In the left sidebar, under Instances, click Instances

Step 5: On the Instances page, click the orange Launch instances button.



Step 7: You will be taken to the Launch an instance page. Under the Quick Start tab, you will see several Amazon Machine Images (AMIs). The default selected is Amazon Linux 2023 AMI (kernel-6.1) – Free tier eligible.



Step 8: Scroll down to the Key pair (login) section. Click Create a new key pair.

Step 9: Click the orange Launch instance button

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

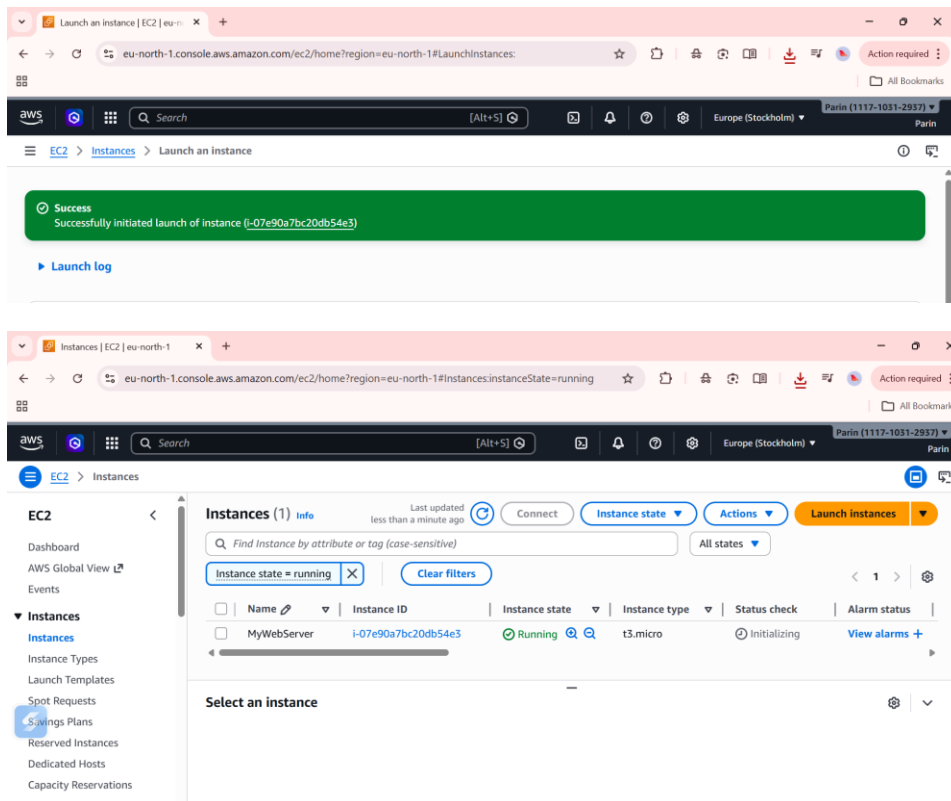
☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

[Cancel](#) [Launch instance](#)

Step 10: A green Success banner will appear: 'Successfully initiated launch of instance'.

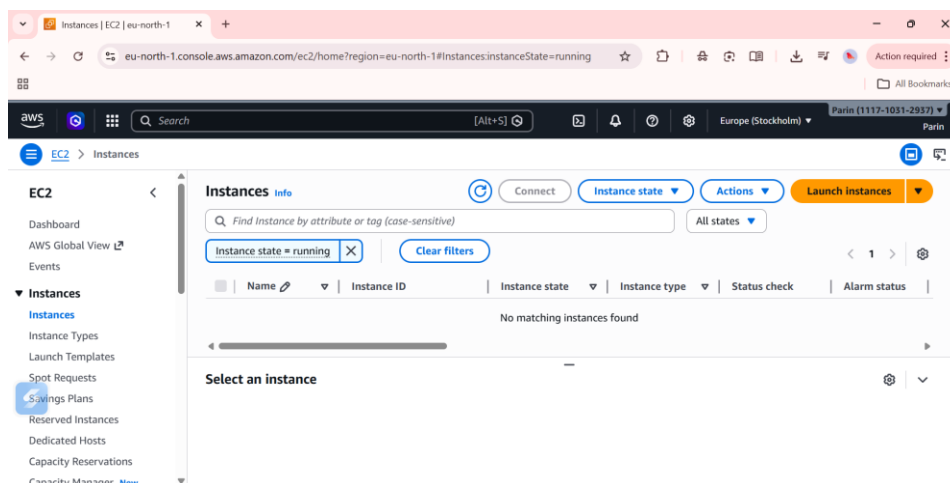
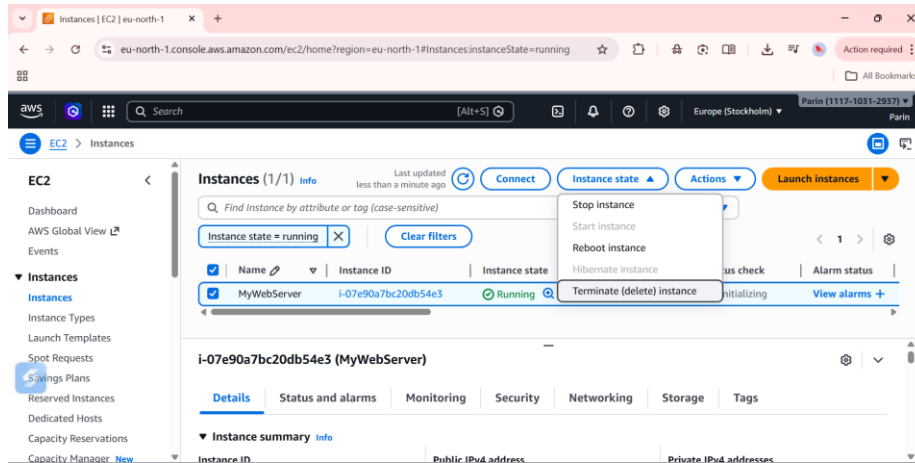


Step 11: In the Instances list, check the checkbox next to MyWebServer to select it.

Step 12: Click the Instance state dropdown button in the top toolbar.

Step 13: A dropdown menu will appear with options: Stop instance, Start instance, Reboot instance, Hibernate instance, Terminate (delete) instance. Click Terminate (delete) instance

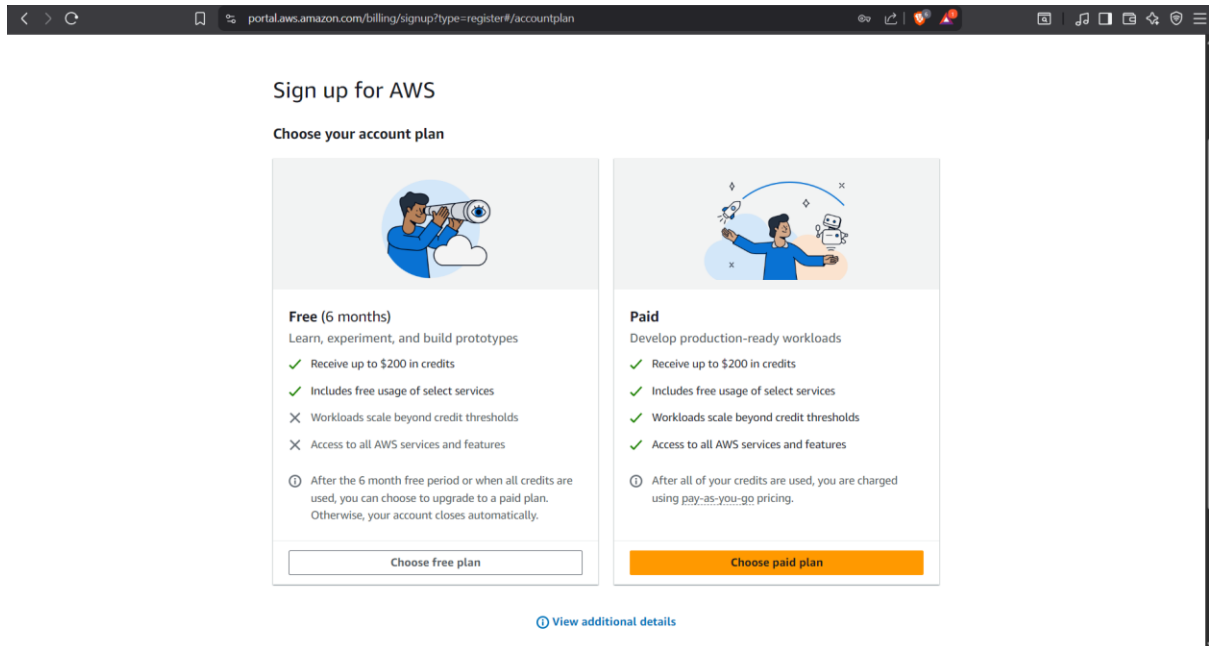
Step 14: A confirmation dialog will appear. Confirm the termination. The instance state will change to Shutting down and then to Terminated.



Practical 4 : Setting up free tier Account AWS & Understand AWS Management

Step 1: Go to AWS Website

1. Open your browser.
2. Visit : <https://aws.amazon.com>
3. Click on **Create an AWS Account** (top-right corner).



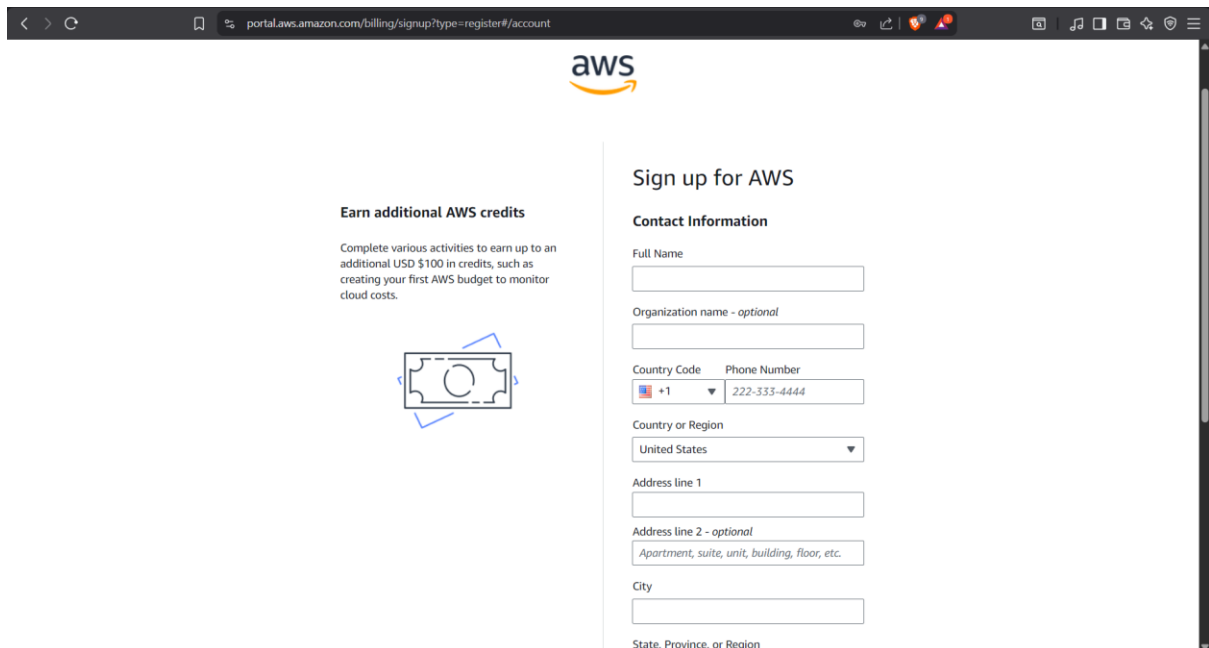
Step 2: Enter Account Details

1. Enter your **email address**.
2. Enter an **AWS account name** (example: Narayan-AWS-Account).
3. Click **Verify email address**.
4. Enter the OTP sent to your email.

Step 4: Contact Information

1. Select **Personal** account type (for students/learning).
2. Enter:
 - Full Name
 - Phone Number
 - Address
 - City, State, Country
3. Accept AWS agreement.

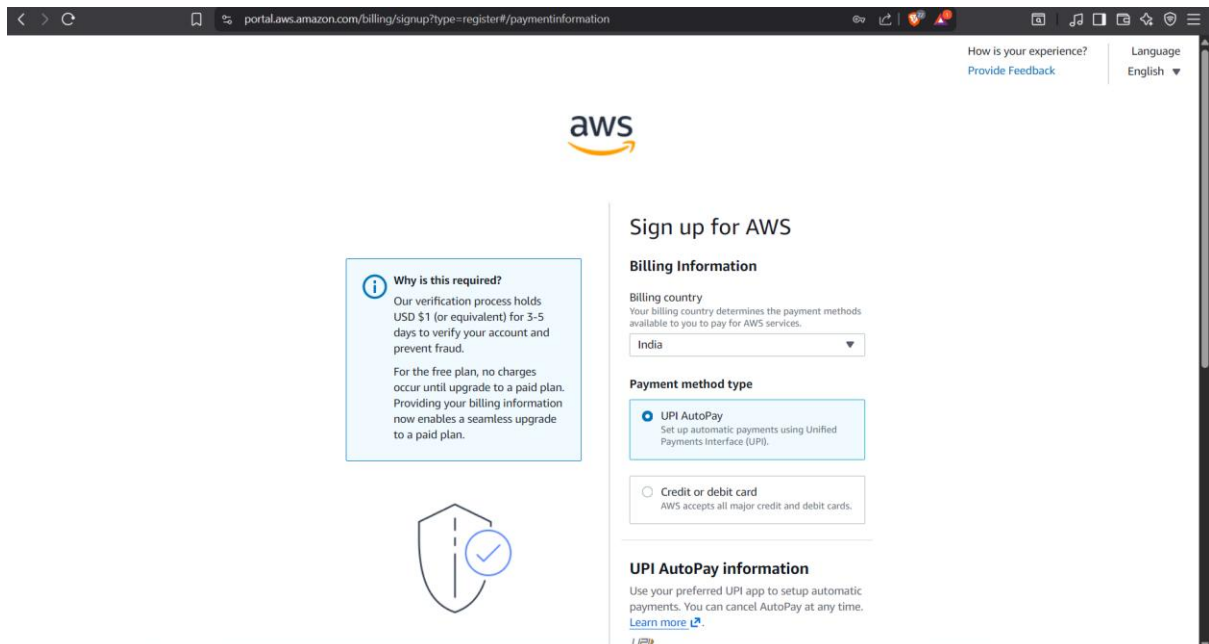
4. Click **Continue**.



The screenshot shows the AWS sign-up page. On the left, there's a section titled "Earn additional AWS credits" with a description: "Complete various activities to earn up to an additional USD \$100 in credits, such as creating your first AWS budget to monitor cloud costs." Below this is an illustration of a credit card. On the right, the "Sign up for AWS" section is active, showing "Contact Information" fields: Full Name, Organization name (optional), Country Code (dropdown with +1 selected), Phone Number (222-333-4444), Country or Region (dropdown with United States selected), Address line 1, Address line 2 (optional, with placeholder text "Apartment, suite, unit, building, floor, etc."), City, and State, Province, or Region.

Step 5: Add Payment Information

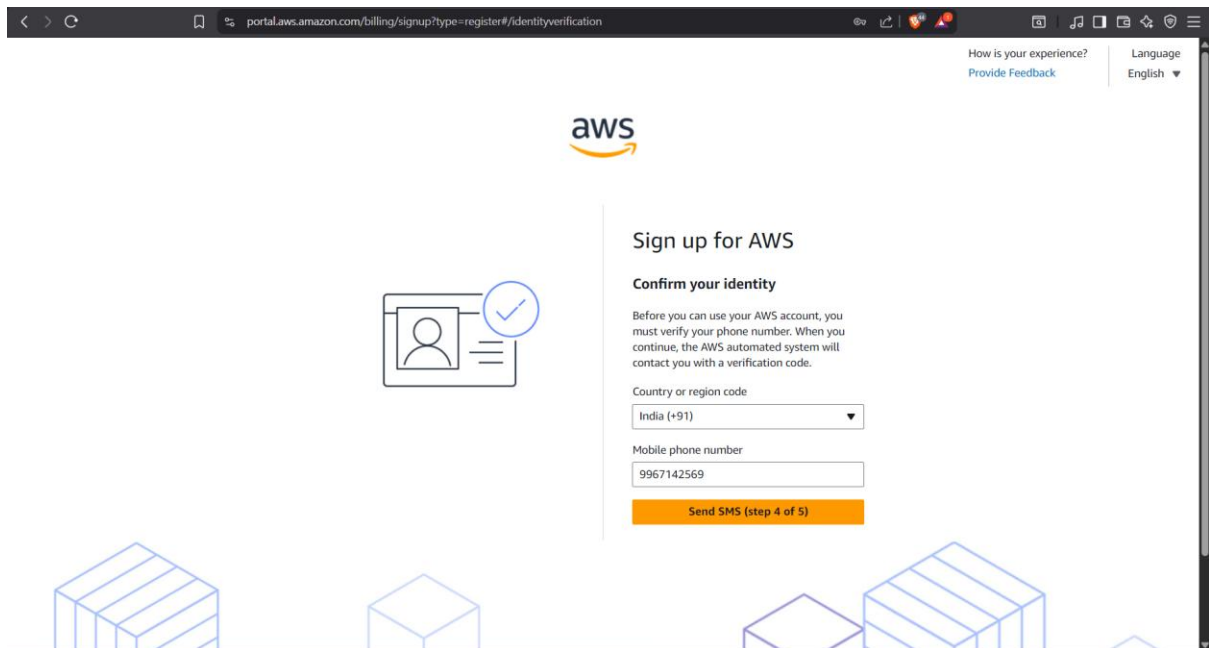
1. Enter your **Debit/Credit card details**.
2. AWS may charge a small temporary amount (₹2–₹5 approx) for verification.
3. Click **Verify and Continue**.



The screenshot shows the AWS sign-up page at the "Billing Information" step. On the left, a blue box titled "Why is this required?" explains: "Our verification process holds USD \$1 (or equivalent) for 3-5 days to verify your account and prevent fraud. For the free plan, no charges occur until upgrade to a paid plan. Providing your billing information now enables a seamless upgrade to a paid plan." Below this is a shield icon with a checkmark. On the right, the "Sign up for AWS" section shows "Billing Information" fields: Billing country (dropdown with India selected) and Payment method type (radio buttons for UPI AutoPay and Credit or debit card). The UPI AutoPay option is selected. Below this is a section for "UPI AutoPay information" with a link to "Learn more".

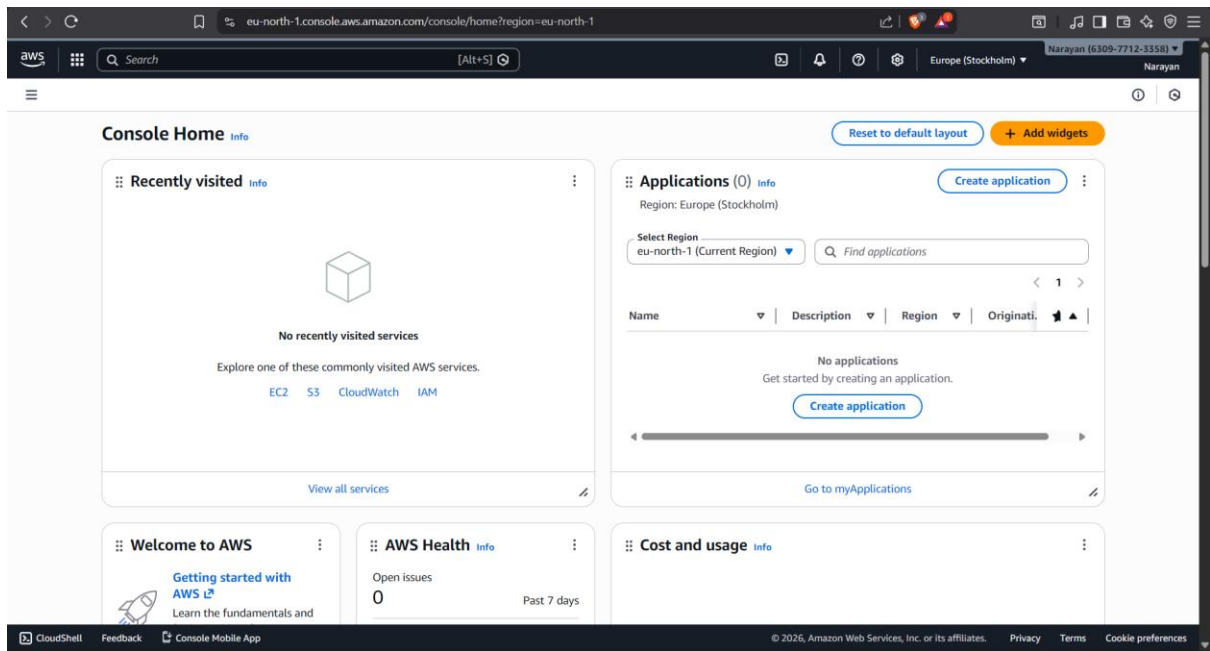
Step 6: Identity Verification

1. Enter your phone number.
2. Choose SMS or Voice call.
3. Enter the OTP received.

A screenshot of the AWS Identity Verification page in a web browser. The browser's address bar shows the URL: portal.aws.amazon.com/billing/signup?type=register#/identityverification. The page features the AWS logo at the top center. Below it, on the left, is an icon of a document with a checkmark. On the right, the heading "Sign up for AWS" is followed by the subheading "Confirm your identity". A paragraph explains that a phone number must be verified and that an automated system will provide a verification code. Below this, there is a dropdown menu for "Country or region code" set to "India (+91)", a text input field for "Mobile phone number" containing "9967142569", and an orange button labeled "Send SMS (step 4 of 5)". The top right corner includes a "Provide Feedback" link and a language selector set to "English". The bottom of the page has a decorative background of blue wireframe cubes.

Step 8: Login to AWS Console

1. Go to : <https://aws.amazon.com/console>
2. Click **Sign in to the Console**.
3. Choose **Root user**.
4. Enter email and password.

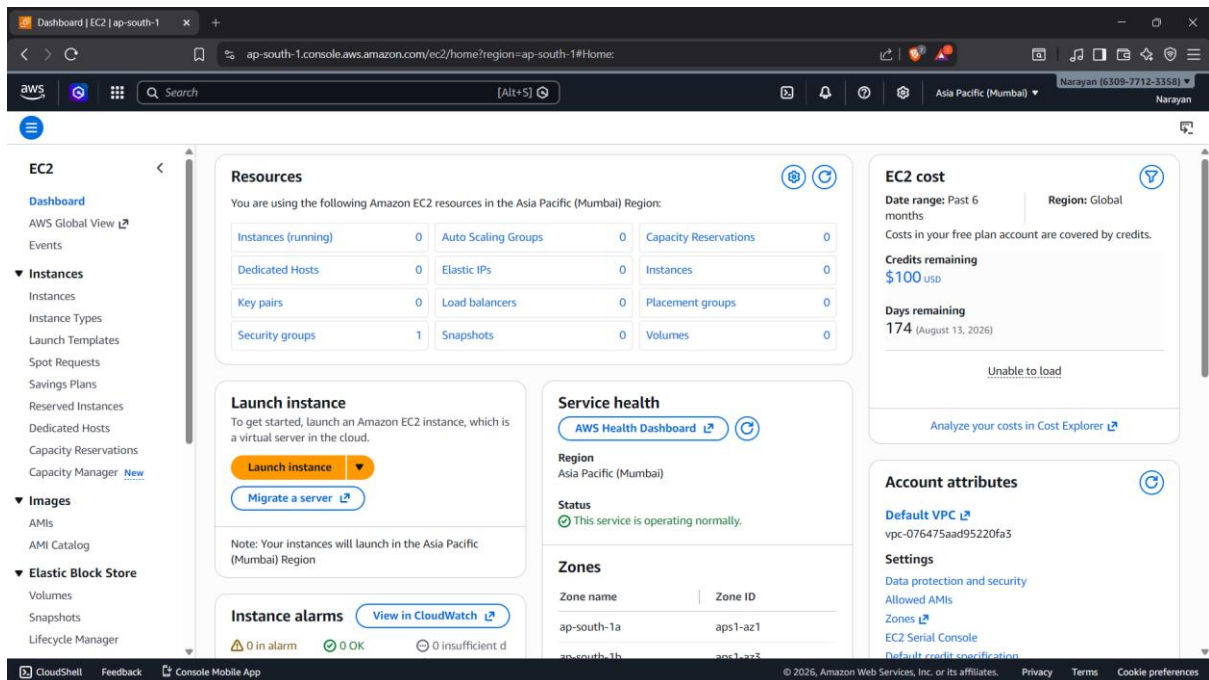


Practical 6 : Concept and LAB of elastic compute storage in AWS

◆ Steps to Create an Instance on Amazon Web Services Amazon EC2

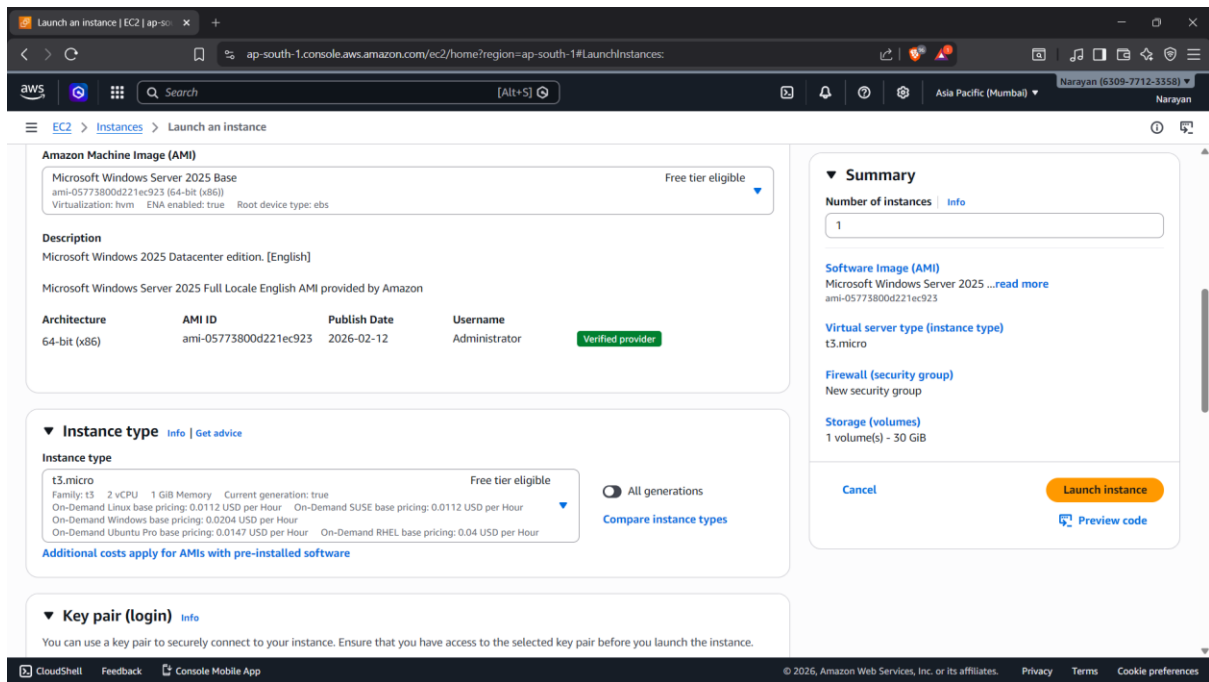
1 Login to AWS

- Go to <https://aws.amazon.com>
- Click **Sign In to Console**
- Enter your credentials



2 Open EC2 Dashboard

- In the search bar, type **EC2**
- Click on **EC2** under Services
- Click **Launch Instance**

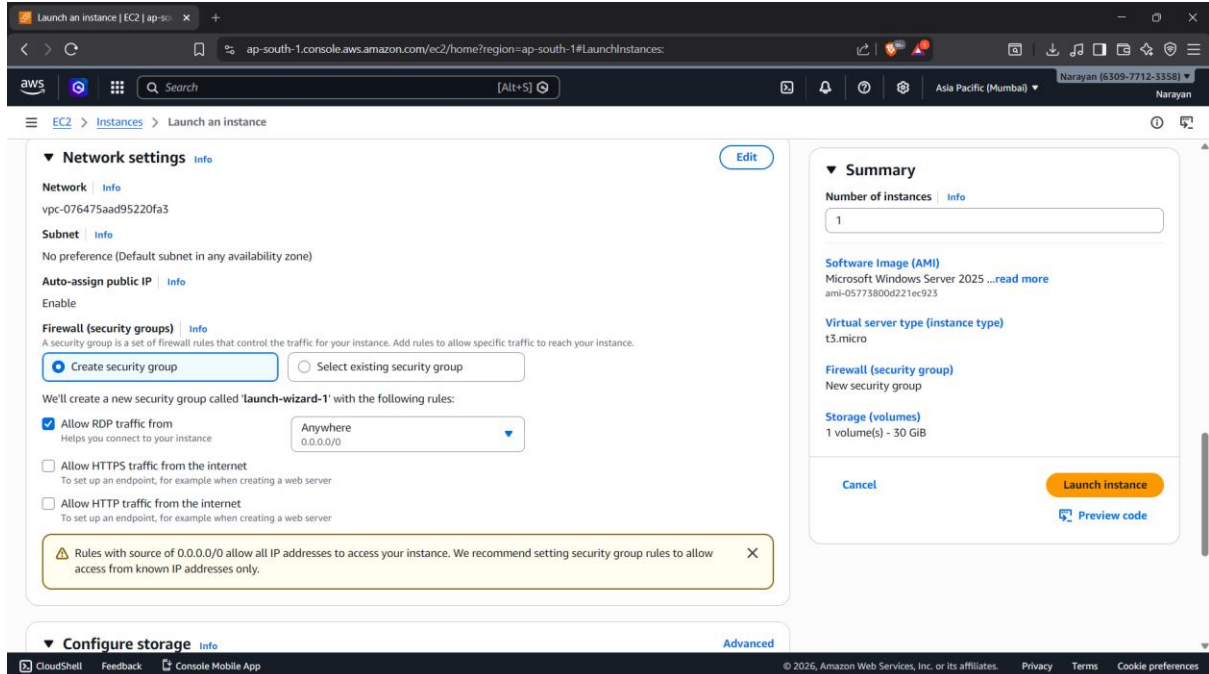


4 Choose Instance Type

- Select instance type

Example:

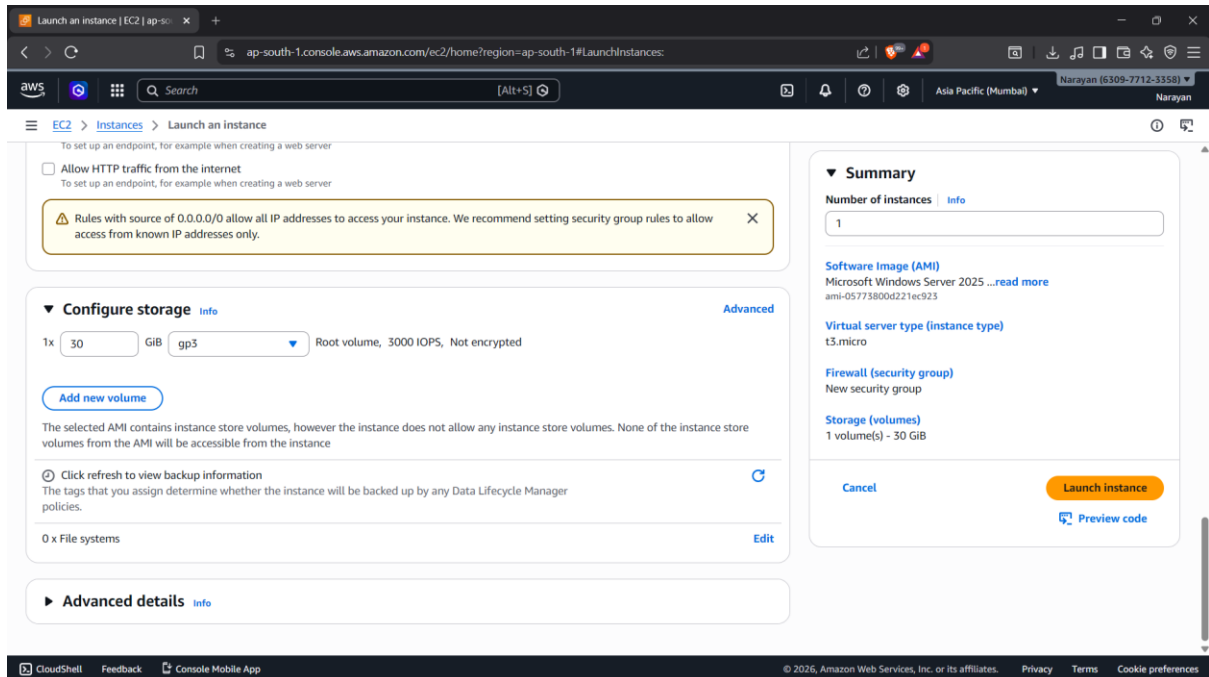
- t2.micro (Free Tier eligible)



5 Create / Select Key Pair

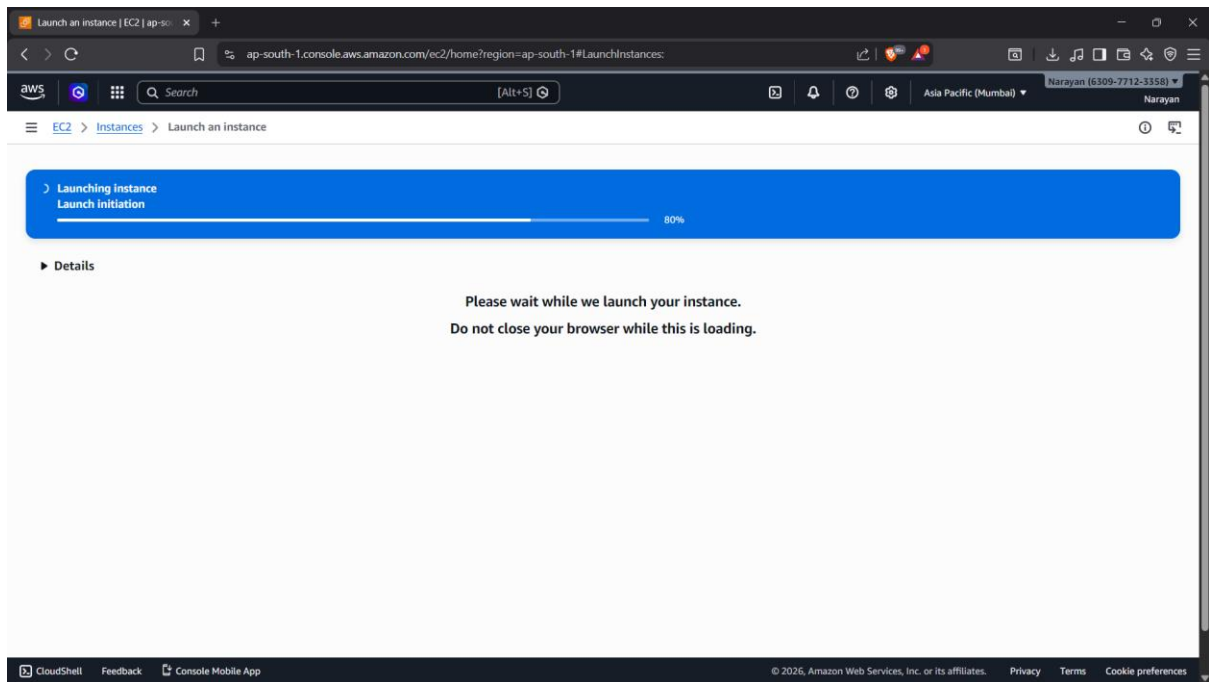
- Click Create new key pair
- Give name
- Choose:

- **RSA**
- **.pem (for Linux/Mac)**
- **.ppk (for Windows)**
- **Download and save it safely**



6 Configure Network Settings

- **Allow required ports:**
 - **SSH (Port 22)**
 - **HTTP (Port 80)**
 - **HTTPS (Port 443)**
- **Select or keep default VPC**



8 Review & Launch

- Click Launch Instance
- Wait for instance state → Running

Launch an instance | EC2 | ap-south-1

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:

Search [Alt+S]

Asia Pacific (Mumbai) Narayan (5309-7712-3358) Narayan

EC2 > Instances > Launch an instance

Success

Successfully initiated launch of instance (i-0cec0a10d7c807bb0)

Launch log

Initializing requests

Succeeded

Creating security groups

Succeeded

Creating security group rules

Succeeded

Launch initiation

Succeeded

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

1

2

3

4

5

6

Create billing usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

Learn more

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

Connect an RDS database

Create a new RDS database

Learn more

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

CloudShell

Feedback

Console Mobile App

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Instances | EC2 | ap-south-1

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#Instances:

Search [Alt+S]

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EC2 > Instances

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Instances (1) Info

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

Name

Instance ID

Instance state

Instance type

Status check

Alarm status

Availability Zone

Public IPv4

myfirstserver

i-0cec0a10d7c807bb0

Running

t3.micro

3/3 checks passed

View alarms

ap-south-1a

ec2-13-201

Select an instance

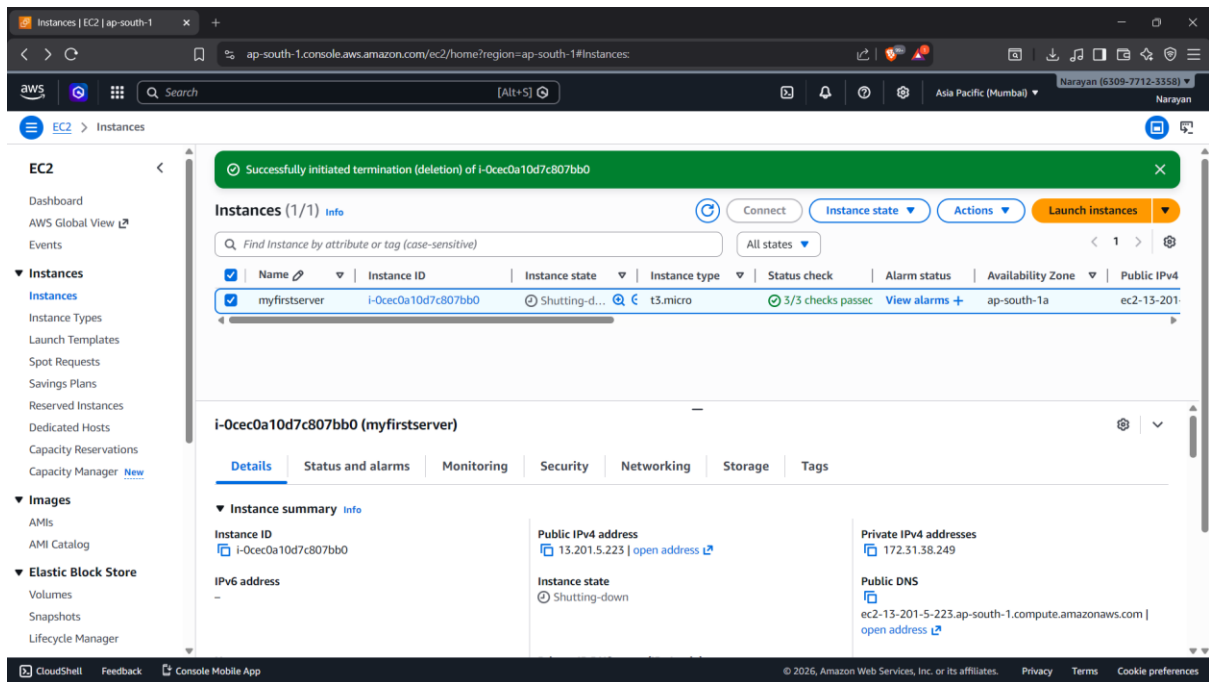
CloudShell

Feedback

Console Mobile App

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3 Choose Name & AMI

- Enter Instance Name
- Choose an Amazon Machine Image (AMI)

Example:

- Amazon Linux
- Ubuntu
- Windows Server

